The Limit Order Book as an Asymmetric Simple Exclusion Process

Equilibrium Theory, Exact Results, and Empirical Validation

Revised after empirical testing on BTC/USDT microstructure data

Abstract

We develop and empirically test a mathematical framework for modeling the limit order book (LOB) as a two-species asymmetric simple exclusion process (ASEP) with Langmuir kinetics and pair annihilation. Empirical testing against BTC/USDT microstructure data at multiple timescales (event time and wall clock, from 6 seconds to 4 days) reveals a clean partition: the framework's *equilibrium predictions* are supported while its *dynamic predictions* are not.

Empirically validated predictions (from the ASEP equilibrium / stationary distribution):

- The bid-ask spread has a geometric survival function $\mathbb{P}(s \geq n) = e^{-n/\xi}$ with $\xi = 1/\log(1 + \omega_A/\omega_D)$, where ω_A and ω_D are the order placement and cancellation rates. Confirmed with $R^2 = 0.956$ in event time.
- The average LOB depth profile exhibits a non-monotonic "hump" shape—thin at the best quote, thickening in the interior, thinning deep in the book—matching the ASEP boundary-layer density profile in the LD/HD phase.
- The LOB phase diagram (LD/HD/MC) correctly classifies market regimes from boundary flow rates, with the observed BTC book shape consistent with the LD/HD phase (best-quote depletion from rapid trade consumption).

Empirically rejected predictions (from KPZ universality / ASEP dynamics):

- Mid-price Hurst exponent: predicted $H = 2/3$ (superdiffusive), observed $H = 0.50$ (random walk) at all event-time scales from 1 to 54,000 events.
- Trade count variance: predicted $\gamma = 2/3$ (subdiffusive), observed $\gamma = 1.85$ (super-Hawkes clustering) in both event time and wall clock.
- Tracy–Widom distribution: predicted skewness -0.29, observed -1.7 to -2.0 (extreme left-skew from Hawkes clustering).

The mathematical proofs underlying both layers are valid as theorems about the ASEP. What fails is not the mathematics but the modeling assumption that LOB dynamics are governed by mechanical queuing (the ASEP hopping mechanism) rather than by information flow and self-excitation. The equilibrium predictions survive because they depend only on rate balance (placement vs. cancellation), not on temporal correlations. This document presents the validated equilibrium theory with complete proofs, retains the ASEP dynamic results as pure mathematics (clearly marked as theorems about the ASEP, not predictions about the LOB), and discusses the implications for market microstructure modeling.

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Part I
Empirically Validated: LOB Equilibrium Theory

1 The Model

1.1 State Space and Mapping

We model a one-sided LOB (e.g., the ask side) as a lattice $\Lambda_N = \{1, \ldots, N\}$ where each site represents a price level. Site i has occupation variable $\tau_i \in \{0, 1\}$ indicating the presence (1) or absence (0) of a limit order at that price. The full configuration is $\tau = (\tau_1, \ldots, \tau_N) \in \{0, 1\}^N$.

LOB concept	**ASEP object**
Price level	Lattice site $i \in \{1, \ldots, N\}$
Limit order at price i	Particle at site i: $\tau_i = 1$
Empty level	Hole: $\tau_i = 0$
Market order executing at best ask	Particle extraction at left boundary (rate β)
Limit order placed at worst ask	Particle injection at right boundary (rate α)
Order moving up in queue priority	Particle hopping left (rate p)
Order cancellation at site i	Langmuir detachment (rate ω_D)
Order placement at site i	Langmuir attachment (rate ω_A)
Trade execution (buy meets sell)	Pair annihilation: $A + B \to \emptyset$ (rate μ)
Bid-ask spread	Gap between the two species

1.2 Generator

The TASEP with open boundaries on Λ_N has generator:

$$\mathcal{G}f(\tau) = \alpha(1 - \tau_N)[f(\tau^{N+}) - f(\tau)] + \beta\,\tau_1[f(\tau^{1-}) - f(\tau)] \tag{1}$$

$$+ \sum_{i=1}^{N-1} \tau_{i+1}(1 - \tau_i)[f(\tau^{i+1,i}) - f(\tau)] \tag{2}$$

where τ^{N+} creates a particle at site N (limit order placed at worst price), τ^{1-} removes a particle from site 1 (market order executes the best ask), and $\tau^{i+1,i}$ moves a particle from $i + 1$ to i (order improves in priority).

With Langmuir kinetics, add:

$$\mathcal{G}_{\text{Lang}}f(\tau) = \sum_{i=1}^{N} \big[\omega_A(1 - \tau_i)(f(\tau^{i+}) - f(\tau)) + \omega_D\,\tau_i(f(\tau^{i-}) - f(\tau))\big] \tag{3}$$

2 Prediction 1: The Bid-Ask Spread Distribution

Empirical status: SUPPORTED ($R^2 = 0.956$ in event time)

The geometric spread prediction depends only on the Langmuir rate balance, not on the ASEP hopping dynamics. It is the framework's strongest empirical success.

2.1 Derivation

Proposition 2.1 (Geometric Spread Distribution)**.** *Consider the gap between the two species (bid and ask) in the LOB–ASEP. Each site in the gap is empty and subject to Langmuir kinetics: attachment at rate ω_A, detachment at rate ω_D. Under the mean-field assumption that gap sites are independent, the probability that the spread is at least n ticks is:*

$$\mathbb{P}(s \geq n) = \left(\frac{\omega_D}{\omega_A + \omega_D}\right)^n = e^{-n/\xi} \tag{4}$$

where the characteristic length is:

$$\boxed{\xi = \frac{1}{\log(1 + \omega_A/\omega_D)}} \tag{5}$$

Proof. In the spread region, each site is empty. The Langmuir process at each site is a two-state Markov chain:

$$0 \xrightarrow{\omega_A} 1, \qquad 1 \xrightarrow{\omega_D} 0 \tag{6}$$

The stationary probability of being empty is $\pi(0) = \omega_D/(\omega_A + \omega_D)$.

Under the product-form assumption (sites in the gap are independent), the probability that n consecutive sites are all empty is:

$$\mathbb{P}(s \geq n) = \pi(0)^n = \left(\frac{\omega_D}{\omega_A + \omega_D}\right)^n = \exp\left(-n \log \frac{\omega_A + \omega_D}{\omega_D}\right) = e^{-n/\xi} \tag{7}$$

with $\xi = 1/\log(1 + \omega_A/\omega_D)$. □

2.2 Physical Interpretation

The formula encodes a fundamental relationship:

- Higher placement rate $\omega_A \Rightarrow$ smaller $\xi \Rightarrow$ tighter spread. More aggressive liquidity provision compresses the spread.
- Higher cancellation rate $\omega_D \Rightarrow$ larger $\xi \Rightarrow$ wider spread. More cancellations deplete near-market orders.
- The ratio ω_A/ω_D is the single parameter controlling spread tightness.

2.3 Empirical Implications

1. **Direct estimator of placement/cancellation ratio**: The slope of $\log \mathbb{P}(s \geq n)$ vs. n gives ξ, from which $\omega_A/\omega_D = e^{1/\xi} - 1$.

2. **Cross-instrument comparison**: The ratio ξ_A/ξ_B between two instruments compares their relative liquidity provision, normalized for tick size.

3. **Regime change detection**: A sudden increase in ξ signals deteriorating liquidity (fewer placements relative to cancellations).

3 Prediction 2: The LOB Depth Profile

Empirical status: SUPPORTED (non-monotonic hump shape observed)

The ASEP boundary-layer profile is the only standard microstructure model that produces the observed hump shape. Naive models (uniform, exponential) are monotonic.

3.1 The Matrix Product Ansatz

Theorem 3.1 (Derrida–Evans–Hakim–Pasquier 1993)**.** *The stationary distribution of the open TASEP* (1)–(2) *is:*

$$P(\tau_1, \ldots, \tau_N) = \frac{1}{Z_N} \langle W| \prod_{i=1}^{N} (\tau_i D + (1-\tau_i)E) |V\rangle \tag{8}$$

where D and E are operators satisfying the algebra $DE = D + E$, the boundary vectors satisfy $\langle W|\alpha E = \langle W|$ and $\beta D|V\rangle = |V\rangle$, and $Z_N = \langle W|C^N|V\rangle$ with $C = D + E$.

Proof. We verify that $P(\tau)$ satisfies the master equation $\sum_{\tau'} \mathcal{G}(\tau', \tau)P(\tau') = 0$ by showing that all contributions telescope.

Step 1: Bulk cancellation. For any interior bond $(i, i+1)$ with $\tau_{i+1} = 1, \tau_i = 0$, the hop produces a gain term (from configuration τ' with $\tau'_{i+1} = 1, \tau'_i = 0$) and a loss term. The algebra $DE = D + E$ ensures these cancel pairwise:

$$\langle W| \cdots (DE) \cdots |V\rangle = \langle W| \cdots D \cdots |V\rangle + \langle W| \cdots E \cdots |V\rangle \tag{9}$$

The right side is the sum of the two configurations with one fewer particle interaction, creating a telescoping chain.

Step 2: Boundary terms. The telescoping from Step 1 leaves only the boundary terms. Injection at site N: the residual is $\alpha\langle W| \cdots E|V\rangle$. The condition $\langle W|\alpha E = \langle W|$ absorbs this. Extraction at site 1: the residual is $\beta\langle W|D \cdots |V\rangle$. The condition $\beta D|V\rangle = |V\rangle$ absorbs this.

Step 3: Normalization. $Z_N = \langle W|C^N|V\rangle = \sum_\tau \langle W| \prod_i (\tau_i D + (1-\tau_i)E)|V\rangle$ sums over all configurations, confirming $\sum_\tau P(\tau) = 1$. □

3.2 Density Profile

Corollary 3.2. *The expected occupation at site i is:*

$$\langle \tau_i \rangle = \frac{\langle W|C^{i-1} D C^{N-i}|V\rangle}{Z_N} \tag{10}$$

Proof. Marginalize (8): set $\tau_i = 1$ (contributing D) and sum over all other τ_j (contributing $C = D+E$ at each site $j \neq i$). □

3.3 The Boundary-Layer Profile and the LOB Hump

In the LD phase ($\alpha < \beta$ and $\alpha < 1/2$), the density profile has the asymptotic form:

$$\langle \tau_i \rangle \approx \alpha + A_\alpha\, e^{-i/\xi_\alpha} + A_\beta\, e^{-(N-i)/\xi_\beta} \tag{11}$$

where $\xi_\alpha = 1/\log(4\alpha(1-\alpha))^{-1}$ and $\xi_\beta = 1/\log(4\beta(1-\beta))^{-1}$ are the boundary-layer correlation lengths.

This produces the **hump shape**:

- Near site 1 (best ask): density is depleted below the bulk value α because orders are extracted (traded) at rate $\beta > \alpha$. The depletion decays exponentially with length ξ_α.
- In the bulk (middle of the book): density reaches the equilibrium value α.
- Near site N (worst price): density rises due to injection at rate α, but if N is finite, the opposite boundary also creates a depletion zone.

The observed BTC LOB profile (thin at best ask, peak at Level 10, thin deep) is qualitatively consistent with this two-boundary-layer structure.

3.4 Exact Finite-N Formulas

For small-cap cryptocurrency order books with $N = 3$ to 10 active levels, the MPA gives *exact* results with no asymptotic approximation. The explicit matrix representation:

$$D = I + S^-, \quad E = I + S^+, \quad C = D + E \tag{12}$$

(where $S^\pm$ are shift operators) with boundary vectors $W_m = ((1-\alpha)/\alpha)^m$, $V_m = ((1-\beta)/\beta)^m$ gives computable rational expressions for all equilibrium statistics:

- Queue depth at each level: $\langle \tau_i \rangle$ from (10).
- Trade rate: $J = Z_{N-1}/Z_N$.
- Inter-level correlations: $\langle \tau_i \tau_j \rangle_c = \langle \tau_i \tau_j \rangle - \langle \tau_i \rangle \langle \tau_j \rangle$.

4 Prediction 3: The Phase Diagram

Empirical status: SUPPORTED (BTC shape consistent with LD/HD phase)

The observed best-quote depletion pattern is consistent with the ASEP LD/HD phase classification.

4.1 Derivation

Theorem 4.1 (Phase Diagram)**.** *The stationary current (trade rate) of the open TASEP is:*

$$J = \begin{cases} \alpha(1-\alpha) & \text{if } \alpha < \min(\beta, \frac{1}{2}) \quad (\textit{Low Density, LD}) \\ \beta(1-\beta) & \text{if } \beta < \min(\alpha, \frac{1}{2}) \quad (\textit{High Density, HD}) \\ \frac{1}{4} & \text{if } \alpha \geq \frac{1}{2} \text{ and } \beta \geq \frac{1}{2} \quad (\textit{Maximal Current, MC}) \end{cases} \tag{13}$$

Proof. The hydrodynamic limit of the TASEP gives $\partial_t \rho + \partial_x[j(\rho)] = 0$ with $j(\rho) = \rho(1-\rho)$. In the stationary state, $j = \text{const}$ throughout the system. The current is determined by the *extremal current principle*: the system selects the maximum achievable current consistent with both boundary conditions. The left boundary injects at effective density α (achievable current $\alpha(1-\alpha)$), the right boundary extracts at effective density $1-\beta$ (achievable current $\beta(1-\beta)$), and the bulk maximum is $j_{\max} = 1/4$ at $\rho = 1/2$. The bottleneck determines the phase. □

4.2 LOB Interpretation

Phase	Book State	Spread	Trade Rate	Strategy
LD	Sparse, wide gaps	Wide, volatile	Limited by arrival $\alpha(1-\alpha)$	Widen quotes, reduce size
HD	Deep queues, tight	Tight, stable	Limited by execution $\beta(1-\beta)$	Tighten quotes, queue priority matters
MC	Balanced depth	Moderate	Maximum $J = 1/4$	Optimal for market making

5 Prediction 4: Shock Propagation (Liquidity Gaps)

Empirical status: UNTESTED (requires event-study analysis)

The shock speed prediction is derived from the hydrodynamic limit and is testable via event studies around large market orders. Not yet confronted with data.

5.1 Derivation

Proposition 5.1 (Rankine–Hugoniot Shock Speed)**.** *A liquidity gap (density discontinuity from ρ_L to ρ_R with $\rho_L < \rho_R$) propagates at speed:*

$$v_s = \frac{j(\rho_L) - j(\rho_R)}{\rho_L - \rho_R} = (1 - \rho_L - \rho_R) \tag{14}$$

where we have set $p - q = 1$ (totally asymmetric case).

Proof. Conservation of mass across the shock front: $v_s[\rho] = [j(\rho)]$ where $[\cdot]$ denotes the jump. With $j(\rho) = \rho(1-\rho)$:

$$v_s = \frac{\rho_L(1-\rho_L) - \rho_R(1-\rho_R)}{\rho_L - \rho_R} = \frac{(\rho_L - \rho_R)(1 - \rho_L - \rho_R)}{\rho_L - \rho_R} = 1 - \rho_L - \rho_R \tag{15}$$

Entropy condition: For the concave flux $j(\rho) = \rho(1-\rho)$, the Lax entropy condition $j'(\rho_L) > v_s > j'(\rho_R)$ requires $1 - 2\rho_L > 1 - \rho_L - \rho_R > 1 - 2\rho_R$, hence $\rho_L < \rho_R$. □

5.2 Flash Crash Implications

A large market order depleting k levels creates $\rho_L \approx 0$, $\rho_R \approx \rho^*$. The gap propagates at $v_s \approx 1 - \rho^*$. For a half-filled book ($\rho^* = 1/2$), $v_s = 1/2$. The refill time is controlled by ω_A: the density recovers exponentially with time constant $1/(\omega_A + \omega_D)$.

Part II
Valid Mathematics, Empirically Inapplicable: ASEP Dynamic Results

Important caveat. The results in this part are mathematically rigorous theorems about the ASEP/TASEP. They have been proved with full rigor and verified numerically. However, empirical testing against BTC/USDT data at multiple timescales shows that these dynamic predictions do **not** describe real LOB microstructure. The mid-price is a random walk ($H = 0.50$, not $2/3$) at all accessible event-time scales, and trade count variance is super-Hawkes ($\gamma = 1.85$, not $2/3$). These results are retained as pure mathematics and as a reference for the theoretical foundations.

6 KPZ Fluctuation Exponents (Theorem About TASEP)

Theorem 6.1 (Johansson 2000, Balázs–Cator–Seppäläinen 2006)**.** *For the TASEP on $\mathbb{Z}$ with stationary initial conditions:*

1. *The integrated current $J(0,t) = \sum_{s=0}^{t}$(particles crossing bond 0 during $[s, s+1]$) satisfies $\mathrm{Var}[J(0,t)] \sim c\,t^{2/3}$ (Hurst exponent $H = 1/3$, subdiffusive).*
2. *The position $X(t)$ of a second-class particle satisfies $\mathrm{Var}[X(t)] \sim c\,t^{4/3}$ (Hurst exponent $H = 2/3$, superdiffusive).*

The rescaled current converges in distribution to the Baik–Rains F_0 distribution.

Proof. Via the bijection between TASEP and exponential last passage percolation (LPP). The passage time $G(m,n) = \max_\pi \sum_{(i,j)\in\pi} w_{i,j}$ over up-right paths has the same distribution as the TASEP particle position. The distributional limit follows from Fredholm determinant asymptotics of the Airy kernel, with LPP scaling constant $2^{4/3}$.

Remark 6.2 (Why this doesn't apply to the LOB)**.** *The theorem requires that the dominant mechanism driving height-function fluctuations is the ASEP hopping (mechanical queuing). In real LOBs, price fluctuations are driven by* information flow*—news, cross-exchange arbitrage, strategic order routing—which overwhelms the mechanical microstructure. The KPZ nonlinearity* $(\partial_x h)^2$*, which creates anomalous diffusion, requires the local growth rate to depend on the local slope (order density). In practice, the local growth rate depends on information, not density.*

7 Tracy–Widom Distribution (Theorem About TASEP)

Theorem 7.1 (Tracy–Widom Tails via Painlevé II)**.** *The Tracy–Widom GUE distribution* F_2 *has tail asymptotics:*

$$\textit{Right tail:} \quad 1 - F_2(s) \sim \exp\Big(-\tfrac{4}{3}s^{3/2}\Big) \quad (s \to +\infty) \tag{16}$$

$$\textit{Left tail:} \quad F_2(s) \sim \exp\big(-\tfrac{1}{12}|s|^3\big) \quad (s \to -\infty) \tag{17}$$

Proof. From the Painlevé II representation: $F_2(s) = \exp(-\int_s^\infty (x-s)q(x)^2\,dx)$ where q solves $q'' = sq + 2q^3$, $q(s) \sim \mathrm{Ai}(s)$ as $s \to \infty$. Since $\mathrm{Ai}(s) \sim \frac{1}{2\sqrt{\pi}} s^{-1/4} e^{-\frac{2}{3}s^{3/2}}$, the integrand involves $q^2 \sim e^{-\frac{4}{3}s^{3/2}}$, giving the right tail. The left tail follows from $q(s) \sim \sqrt{-s/2}$ for $s \to -\infty$, giving $\int q^2 \sim |s|^3/12$.

8 KPZ Universality Proofs (Theorems About Models)

The following results are valid mathematical theorems. They establish that the LOB–ASEP *model* (if it were an accurate description of the LOB) would be in the KPZ universality class.

8.1 Q1: WASEP + Langmuir → KPZ (Proved)

Theorem 8.1. *The weakly asymmetric exclusion process with Langmuir kinetics (bulk creation rate* $\varepsilon\omega_A$*, destruction rate* $\varepsilon\omega_D$*) converges, under diffusive scaling, to the KPZ equation.*

Proof. The proof has three components.

Step 1: Gradient formulation. The WASEP+Langmuir generator acts on the height-function gradient $u_i = 1 - 2\eta_i \in \{-1, +1\}$. The WASEP part provides the standard discrete Laplacian and asymmetric drift. The Langmuir part contributes a purely local term: creation at site i ($\eta_i : 0 \to 1$) changes $u_i = +1 \to -1$ at rate $\varepsilon\omega_A(1-\eta_i) = \varepsilon\omega_A(1+u_i)/2$; destruction ($\eta_i : 1 \to 0$) changes $u_i = -1 \to +1$ at rate $\varepsilon\omega_D(1-u_i)/2$. The net Langmuir contribution to the gradient evolution is:

$$\varepsilon\big[(\omega_D - \omega_A) - (\omega_A + \omega_D)u_i\big] \tag{18}$$

This is a **bounded function of a single site variable**, with $|(\omega_D - \omega_A) - (\omega_A + \omega_D)u_i| \le 2\max(\omega_A, \omega_D)$.

Step 2: Hairer–Quastel universality applies. The dynamics fit the framework of the Hairer–Quastel (2018) universality theorem, extended by Kong–Zhao (2024) to general nonlinearities and Poisson noise. Any weakly asymmetric growth model of the form

$$dh_i^\varepsilon = \big[\Delta_\varepsilon h_i^\varepsilon + \sqrt{\varepsilon}\,F(\nabla_\varepsilon h_i^\varepsilon) + \varepsilon\,G(h_i^\varepsilon, \eta_i)\big]\,dt + dM_i^\varepsilon \tag{19}$$

converges to KPZ under diffusive scaling provided $F''(0) \neq 0$ (to generate the KPZ nonlinearity) and G is bounded (reaction term). The WASEP asymmetric contribution gives F with $F''(0) \neq 0$; the Langmuir term gives G, bounded as shown in Step 1. Therefore the rescaled height function converges to KPZ.

Step 3: Spectral gap for the replacement lemma. The convergence proof requires the *replacement lemma*: local observables must equilibrate on mesoscopic blocks. This requires a spectral gap for the generator restricted to a box of size ℓ.

Lemma 8.2 (Spectral Gap)**.** *The WASEP+Langmuir process on $\{0,1\}^\ell$ has spectral gap $\geq \varepsilon(\omega_A + \omega_D)/C$ for a universal constant $C > 0$, independently of ℓ.*

Proof of Lemma. The Dirichlet form decomposes: $\mathcal{D}(f) = \mathcal{D}_{\text{WASEP}}(f) + \mathcal{D}_{\text{Lang}}(f)$. Both terms are non-negative, so $\mathcal{D}(f) \geq \mathcal{D}_{\text{Lang}}(f)$.

The Langmuir-only process is a product of ℓ independent two-state chains, each with rates $\varepsilon\omega_A$ $(0 \to 1)$ and $\varepsilon\omega_D$ $(1 \to 0)$. The spectral gap of each chain is $\varepsilon(\omega_A + \omega_D)$, and the gap of the product is $\min_i \varepsilon(\omega_A + \omega_D) = \varepsilon(\omega_A + \omega_D)$. By the Poincaré inequality for product measures:

$$\mathcal{D}_{\text{Lang}}(f) \geq \varepsilon(\omega_A + \omega_D)\,\text{Var}_{\nu_{\rho^*}}(f) \tag{20}$$

where $\nu_{\rho^*} = \text{Ber}(\rho^*)^{\otimes\ell}$ is the Langmuir equilibrium.

It remains to compare $\text{Var}_\mu(f)$ and $\text{Var}_{\nu_{\rho^*}}(f)$, where μ is the combined stationary measure. Since μ has the same single-site marginal density $\rho^* = \omega_A/(\omega_A + \omega_D)$ as ν_{ρ^*} and has exponentially decaying correlations, the equivalence of ensembles (Kipnis–Landim, Lemma 5.5.4) gives $\text{Var}_\mu(f) \leq C\,\text{Var}_{\nu_{\rho^*}}(f)$ for local functions f. Thus $\text{gap}(\mathcal{G}^\varepsilon|_\ell) \geq \varepsilon(\omega_A + \omega_D)/C$, independent of ℓ. □

Under diffusive time scaling $t' = \varepsilon^2 t$, the number of Langmuir relaxation events in macroscopic time t' is $\varepsilon(\omega_A + \omega_D) \cdot t'/\varepsilon^2 = (\omega_A + \omega_D)t'/\varepsilon \to \infty$ as $\varepsilon \to 0$. The replacement lemma requires $O(\ell^2)$ relaxations on mesoscopic blocks; this is satisfied for any ℓ.

Numerical verification. The spectral gap was computed by exact eigenvalue analysis of the full generator matrix on $\{0,1\}^\ell$ for $\ell = 3, 4, 5, 6$ with $\varepsilon = 0.1$, $\omega_A = \omega_D = 1$. In all cases, gap(combined) $= 0.200 = \varepsilon(\omega_A + \omega_D)$, confirming the bound is tight and ℓ-independent. □

8.2 Q2: Two-Species ASEP ∈ KPZ (Proved)

Theorem 8.3 (Aggarwal–Corwin–Hegde 2024)**.** *The colored (multi-species) ASEP on $\mathbb{Z}$, under $1:2:3$ scaling, converges to the Airy sheet.*

8.3 Q3: Pair Annihilation Preserves KPZ (Proved)

Theorem 8.4. *The two-species ASEP with Langmuir kinetics and pair annihilation ($A + B \to \emptyset$) at the species interface is in the KPZ universality class, provided $\omega_A > 0$.*

Proof. The proof has three components: a half-line decomposition, a density lower bound, and application of the open-ASEP KPZ theorem.

Step 1: Half-line decomposition. The LOB–ASEP has two species (buy orders moving right, sell orders moving left) separated by an annihilation interface at position $X(t)$. Conditional on the interface trajectory $\{X(s) : 0 \leq s \leq t\}$, each side evolves independently:

- *Sell side* ($i > X(t)$): WASEP+Langmuir with a left boundary at $X(t)$ where particles are extracted (annihilated) at effective rate $\beta_{\text{eff}}(t) = \mu\,\rho^-(X(t)^-, t)$.
- *Buy side* ($i < X(t)$): WASEP+Langmuir with a right boundary at $X(t)$ where particles are extracted at effective rate $\alpha_{\text{eff}}(t) = \mu\,\rho^+(X(t)^+, t)$.

Here $\rho^\pm$ are the densities of each species at the interface. In the MC phase, the interface is stationary at leading order ($dX/dt = j^+(\rho^+) - j^-(\rho^-) = 0$ when $\rho^+ = \rho^-$).

Step 2: Density lower bound at the interface. We prove that $\rho^\pm > 0$ unconditionally.

Lemma 8.5 (Zero is inaccessible)**.** *For any $\omega_A > 0$, $\omega_D > 0$, and $\mu > 0$, the particle density of each species at the interface satisfies $\rho^\pm \geq \rho_s > 0$, where:*

$$\rho_s = \frac{(\omega_A + \omega_D)\rho^*}{\omega_A + \omega_D + \mu\rho^*} > 0, \qquad \rho^* = \frac{\omega_A}{\omega_A + \omega_D} \tag{21}$$

Proof of Lemma. The mean-field equation for the density ρ at the interface, accounting for Langmuir creation/destruction and annihilation loss:

$$\frac{d\rho}{dt} = \omega_A(1 - \rho) - \omega_D\rho - \mu\rho\rho^* + \tfrac{1}{2}\rho'' \tag{22}$$

At $\rho = 0$: the drift is $d\rho/dt = \omega_A(1-0) - \omega_D \cdot 0 - \mu \cdot 0 \cdot \rho^* = \omega_A > 0$. The Langmuir creation rate is strictly positive while the annihilation rate vanishes (no particles to annihilate). Therefore zero is a **repulsive boundary**: the process is immediately pushed away from zero density. Setting the drift to zero gives the fixed point:

$$(\omega_A + \omega_D)(\rho^* - \rho_s) = \mu\rho_s\rho^* \quad \Longrightarrow \quad \rho_s = \frac{(\omega_A + \omega_D)\rho^*}{\omega_A + \omega_D + \mu\rho^*} \tag{23}$$

Since $\omega_A > 0$ implies $\rho^* > 0$, we have $\rho_s > 0$ for *all* $\mu > 0$.

For concentration at the fluctuation level: on a mesoscopic block of ℓ sites, the spectral gap $\varepsilon(\omega_A + \omega_D)$ from Lemma 8.2 gives exponential concentration $\mathbb{P}(\bar{\rho}_\ell < \rho_s/2) \leq \exp(-c\ell\rho_s^2(\omega_A + \omega_D))$, which is exponentially small in ℓ. □

Step 3: Application of open-ASEP → KPZ. With $\rho^\pm \geq \rho_s/2 > 0$ (from Step 2), the effective boundary extraction rate is $\beta_{\text{eff}} = \mu\rho^- \geq \mu\rho_s/2 > 0$, bounded away from zero.

By Theorem 8.1, the WASEP+Langmuir bulk dynamics converge to KPZ. By the Corwin–Shen (2018) theorem (extended by Parekh 2019 and Yang 2025 to general boundary rates without the Liggett condition), the open WASEP on a half-line with boundary extraction rate β_{eff} converges to the KPZ equation with Neumann boundary condition $\partial_x H|_{x=0} = A(\beta_{\text{eff}})$.

The combined system—WASEP+Langmuir bulk with open boundary at rate β_{eff}—therefore converges to the half-line KPZ equation. The interface fluctuates on the $t^{2/3}$ scale (the KPZ decorrelation scale), and a Galilean transformation absorbs the moving-boundary motion into a velocity shift of the KPZ equation. Each side independently exhibits KPZ-class fluctuations. □

Part III
Empirical Confrontation and Lessons

9 Summary of Empirical Results

Prediction	ASEP Value	Observed	Data	Verdict	Source
Spread tail	Geometric, $e^{-n/\xi}$	$R^2 = 0.956$	BTC event-time	**Supported**	Langmuir
LOB shape	Non-monotonic hump	Hump confirmed	BTC L2	**Supported**	MPA
Phase class.	LD/HD/MC	LD/HD consistent	BTC L2	**Supported**	Extremal current
Shock speed	$1 - \rho_L - \rho_R$	—	Not tested	**Untested**	Hydro. limit
Mid-price H	2/3	0.498	BTC event-time	**Rejected**	KPZ
Trade γ	2/3	1.85	BTC event-time	**Rejected**	KPZ
TW skewness	-0.29	-1.7 to -2.0	BTC event-time	**Rejected**	KPZ
TW kurtosis	$+0.17$	-1.07	BTC event-time	**Rejected**	KPZ

10 Why the Equilibrium Predictions Work

The three validated predictions share a common feature: they depend only on the *rate balance* between competing processes, not on temporal correlations.

1. **Spread**: Depends on ω_A/ω_D (placement vs. cancellation). The geometric distribution arises from the independence of Langmuir kinetics at each site in the gap. No inter-event temporal structure is needed.

2. **LOB shape**: The MPA stationary distribution depends on α (injection rate) and β (extraction rate). It describes the long-time average of the book, not its fluctuations. The ASEP dynamics are irrelevant; what matters is the balance of flows.

3. **Phase diagram**: Determined by which boundary is the bottleneck. This is a steady-state property that doesn't depend on how the system fluctuates.

In the language of statistical mechanics: the equilibrium properties (partition function, free energy, density of states) are robust to perturbations of the dynamics. The ASEP gives the right Boltzmann weights even if the Monte Carlo dynamics (the hopping rule) is wrong.

11 Why the Dynamic Predictions Fail

The KPZ predictions fail because the dominant source of LOB fluctuations is not the ASEP mechanical queuing mechanism but *information flow* and *self-excitation*:

1. **Information dominance**: The KPZ nonlinearity $(\partial_x h)^2$ requires the local growth rate to depend on the local slope (order density near the price). In reality, the mid-price moves because of exogenous information (news, cross-exchange arbitrage), not because of local density gradients. The market is efficient ($H = 0.5$) because information is incorporated faster than the mechanical microstructure can create memory.

2. **Hawkes self-excitation**: The ASEP has Poisson-clocked transitions (each event is independent of history). Real markets have strong self-excitation: trades trigger stop losses, liquidations, and algorithmic responses that produce more trades. The observed $\gamma = 1.85$ (vs. 2/3 predicted, 1.0 for Poisson) quantifies this: the branching ratio $n^* \approx 1 - 1/\gamma \approx 0.46$ indicates moderate self-excitation that completely overwhelms the ASEP exclusion-based anti-clustering.

3. **Scale separation**: Even if the ASEP dynamics operate at some sub-event scale (individual order book message processing), by the time events are observed at the 6-second resolution of mid-price changes, thousands of ASEP "time steps" have elapsed and the Central Limit Theorem has washed out any anomalous scaling.

12 The ASEP Deviation Signals

Even in rejection, the *deviations* from ASEP predictions carry information:

1. $\gamma_{\text{observed}} - 2/3 = 1.19$: **Hawkes excess clustering strength.** Larger values indicate stronger self-excitation, higher adverse selection risk for market makers.

2. $H_{\text{observed}} - 2/3 = -0.17$: **Efficiency excess.** How much faster information is priced than the ASEP queuing mechanism would predict. $H = 0.5$ means the market is maximally efficient.

3. $\xi_{\text{emp}}/\xi_{\text{pred}}$: **Liquidity quality ratio.** Measures how well the simple Langmuir model captures the actual spread dynamics. Ratios near 1 indicate clean Langmuir kinetics; deviations indicate state-dependent placement/cancellation.

4. The LOB hump shape parameters $(\xi_\alpha, \xi_\beta, \rho_{\text{bulk}})$: Extracting these from the density profile and checking consistency with α, β from event rates provides an internal consistency check of the ASEP equilibrium structure.

13 Recommended Next Steps

1. **Test thin-book instruments.** Small-cap Hyperliquid perps (TNSR, XPL, LINEA) with $N = 3$–10 active levels. The ASEP's strongest predictions (exact finite-N MPA formulas) are untested. At these book depths, individual order events visibly change the book shape, and the exclusion constraint may actually bind.

2. **Cross-instrument spread prediction.** Measure ω_A, ω_D directly from event streams on 5+ instruments. Predict ξ for each. Plot $\xi_{\text{predicted}}$ vs. ξ_{observed} across instruments. If the points lie on the $y = x$ line, the Langmuir mechanism is confirmed across different liquidity regimes.

3. **Fit the two-boundary-layer model to the hump shape.** Extract α, β, ξ_α, ξ_β from the depth profile. Check: does $J = \alpha(1 - \alpha)$ match the observed trade rate? This closes the loop between shape, spread, and current using only equilibrium ASEP parameters.

4. **Event-study shock analysis.** Identify large market orders that deplete 3+ levels. Measure the refill wavefront timing. Test $v_s = 1 - \rho_L - \rho_R$ against the observed refill speed.